Matching the Analysis Model to the Data-Generating Process: AR(1) Residual Correlation in the pmsimstats LME Analysis

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Abstract

When the `pmsimstats` data-generating process was updated from compound symmetry to AR(1) within-factor autocorrelation to expand the feasible parameter space, the analysis model's assumption of independent residuals (given a random intercept) became misspecified. This misspecification inflated Type I error rates for the OL and CO designs from the nominal 5% to 13–17%. This paper documents the problem, derives the appropriate analysis model, and validates the fix by switching from `lme4::lmer` to `nlme::lme` with a continuous-time AR(1) residual correlation structure (`corCAR1`).

Architecture scope. The AR(1) autocorrelation structure described here applies to both DGP architectures (A and B), as it governs the within-factor temporal correlation in the covariance matrix regardless of whether the biomarker-treatment interaction is in the covariance or mean structure. The Type I error analysis was conducted under Architecture B (MVN differential correlation). For a comparison of both DGP architectures, see `02-dgp-mean-moderation-vs-mvn.tex`. For implementation details of Architecture A, see `19-mean-moderation-implementation-notes.tex`.

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1 The Problem

1.1 Background

The original `pmsimstats` DGP used compound symmetry (CS) autocorrelation: the correlation between the same factor at any two timepoints was a constant $\rho = 0.8$, regardless of the time gap between them. The analysis model used `lme4::lmer` with a random intercept per participant and no explicit residual correlation structure:

```
Sx ~ bm + t + Dbc + bm:Dbc + (1|ptID)     # varInDb
Sx ~ bm + t + bm:t + (1|ptID)             # OL (no varInDb)
```

Under CS, the random intercept absorbs the constant within-person correlation, and the residuals are approximately independent. Type I error rates were near the nominal 5%.

1.2 The AR(1) Change

To expand the feasible biomarker correlation range (from $c_{bm} \leq 0.25$ to $c_{bm} \leq 0.5$), the DGP was updated to use AR(1) autocorrelation:

$$\mathrm{Cor}(\mathrm{Factor}_{t_i}, \mathrm{Factor}_{t_j}) = \rho^{|t_i - t_j|}$$

where t is measured in weeks. At $\rho = 0.7$, adjacent timepoints 2.5 weeks apart have correlation $0.7^{2.5} = 0.41$, while timepoints 20 weeks apart have correlation $0.7^{20} = 0.0008$.

Unlike CS, AR(1) produces time-varying correlations that the random intercept cannot fully absorb. The residuals after removing the random intercept retain autocorrelation that decays with time, and the `lmer` model (which assumes conditionally independent residuals) underestimates the standard errors of the fixed effects.

1.3 Observed Type I Error Inflation

With the AR(1) DGP and the original `lmer` analysis model, Type I error rates at $c_{bm} = 0$ were:

Table 1: Type I error rates with AR(1) DGP and `lmer` analysis (200 replicates, no censoring)

Design	$N = 35$	$N = 70$
OL	0.13	0.09
OL+BDC	0.09	0.05
CO	**0.17**	**0.13**
N-of-1	0.06	0.08

The CO design showed 17% false positive rate at $N = 35$ (3.4× the nominal rate). OL was similarly inflated. OL+BDC and N-of-1 were closer to nominal.

1.4 Why CO and OL Are Most Affected

CO design. The crossover has two paths: drug-first (Path A: on-on-on-on-off-off-off-off) and placebo-first (Path B: off-off-off-off-on-on-on-on). Under AR(1), the autocorrelation within each path decays with time, creating a correlation gradient across timepoints. The `Dbc` variable (binary

on/off drug) is perfectly confounded with the first vs. second half of the trial. The `bm:Dbc` interaction can absorb spurious signal from the time-dependent correlation structure that the random intercept does not capture.

OL design. All participants are always on drug, so the model uses `bm:t` (biomarker × time) as the interaction term. Under AR(1), the residual correlation decays with time, creating time-dependent heteroscedasticity in the residuals that inflates the `bm:t` standard error estimate downward.

OL+BDC and N-of-1. These designs have more complex path structures with multiple transitions between on-drug and off-drug. The `Dbc` variable is less confounded with linear time, so the AR(1) residual structure has less impact on the interaction estimate.

2 The Solution: `nlme::lme` with `corCAR1`

2.1 Continuous-Time AR(1)

The `nlme` package provides the `corCAR1` correlation structure, which models continuous-time AR(1) residual correlation:

$$\mathrm{Cor}(\varepsilon_{i,t_1}, \varepsilon_{i,t_2}) = \phi^{|t_1 - t_2|}$$

where ϕ is estimated from the data. This directly matches the AR(1) structure used in the DGP (with ϕ corresponding to the DGP's ρ).

2.2 Model Specification

The analysis model was changed from `lme4::lmer`:

```
lmer(Sx ~ bm + t + Dbc + bm:Dbc + (1|ptID),
     data = datamerged)
```

to `nlme::lme`:

```
lme(Sx ~ bm + t + Dbc + bm:Dbc,
    random = ~1|ptID,
    correlation = corCAR1(form = ~t|ptID),
    data = datamerged)
```

The key addition is `correlation = corCAR1(form = ~t|ptID)`, which tells `nlme` to estimate and account for AR(1) residual correlation as a function of the time variable `t`, grouped by participant `ptID`.

2.3 Coefficient Extraction

The `nlme::lme` summary uses a different output format than `lme4::lmer`:

- `lmer`: `summary(fit)$coefficients` with columns `Estimate`, `Std. Error`, `Pr(>|t|)`
- `lme`: `summary(fit)$tTable` with columns `Value`, `Std.Error`, `p-value`

The code was updated to extract from the `tTable` format.

2.4 Error Handling

Two levels of fallback were implemented:

1. If `lme` with `corCAR1` fails to converge (can happen with small sample sizes or degenerate data), the model falls back to `lme` without a correlation structure.
2. If the fallback also fails, the function returns `NA` for all output values.

Additionally, rows with `NA` outcomes (from censoring) are removed before fitting, as `nlme::lme` uses `na.fail` by default.

3 Why This Is the Correct Model

3.1 DGP-Analysis Consistency

A simulation study's power estimates are valid when the analysis model is correctly specified relative to the DGP. When the DGP generates data with AR(1) residual autocorrelation, the analysis model must account for this structure. Failure to do so produces:

- **Underestimated standard errors:** The effective sample size is smaller than the nominal sample size because correlated observations carry less independent information. Ignoring the correlation inflates the apparent precision.
- **Inflated Type I error:** The underestimated standard errors produce test statistics that are too large, leading to excess false positives.
- **Biased power estimates:** Power appears higher than it actually is because the test rejects too often under both the null and alternative hypotheses.

3.2 The Random Intercept Is Not Sufficient

Under compound symmetry, the within-person correlation is constant across all timepoint pairs. A random intercept $u_i \sim N(0, \sigma_u^2)$ induces an equicorrelation structure:

$$\mathrm{Cor}(Y_{i,t_1}, Y_{i,t_2}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} = \text{constant}$$

This exactly matches CS, so the residuals after removing the random effect are independent.

Under AR(1), the within-person correlation varies: $\rho^{|t_1 - t_2|}$. The random intercept can absorb the "average" correlation but leaves a residual correlation pattern:

$$\mathrm{Cor}(\varepsilon_{i,t_1}, \varepsilon_{i,t_2} \mid u_i) = \rho^{|t_1 - t_2|} - \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}$$

This residual correlation is positive for nearby timepoints (where $\rho^{|t_1 - t_2|}$ exceeds the average) and negative for distant timepoints (where it falls below). The `lmer` model treats these residuals as independent, producing incorrect standard errors.

3.3 Why `corCAR1` Is Preferred Over Discrete AR(1)

The timepoints in the trial designs are not equally spaced. For example, the N-of-1 design has timepoints at weeks 4, 8, 9, 10, 11, 12, 16, 20, with gaps ranging from 1 to 4 weeks. The continuous-time AR(1) model (`corCAR1`) handles unequal spacing naturally: $\phi^{|t_1-t_2|}$ where t is in weeks. The discrete AR(1) (`corAR1`) would assume equal spacing between consecutive observations, which is incorrect for these designs.

4 Design-Specific Model Selection

The analysis model varies by design based on the structure of the drug exposure variable:

4.1 Designs with Within-Subject Drug Variation (CO, N-of-1, OL+BDC)

These designs have participants who experience both on-drug and off-drug timepoints. The model tests whether the biomarker predicts differential response to drug vs. no-drug:

$$S_{i,t} = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot t + \beta_3 \cdot D_{bc,t} + \beta_4 \cdot bm_i \cdot D_{bc,t} + u_i + \varepsilon_{i,t} \tag{1}$$

where D_{bc} is the binary drug indicator and β_4 is the biomarker–treatment interaction of interest. $\varepsilon_{i,t}$ follows continuous-time AR(1) with parameter ϕ estimated from the data.

4.2 Open-Label Design (OL)

All participants are always on drug, so there is no within-subject drug variation. The model tests whether the biomarker predicts the rate of improvement over time:

$$S_{i,t} = \beta_0 + \beta_1 \cdot bm_i + \beta_2 \cdot t + \beta_3 \cdot bm_i \cdot t + u_i + \varepsilon_{i,t} \tag{2}$$

where β_3 is the biomarker–time interaction. The AR(1) correlation structure is particularly important here because `bm:t` is directly confounded with the time-dependent residual correlation under AR(1).

5 Implementation

The complete model fitting code:

```
# Build fixed-effects formula
modelbase <- "Sx ~ bm + t"
if (varInDb) {
  modelbase <- paste0(modelbase, " + Dbc + bm:Dbc")
} else {
  modelbase <- paste0(modelbase, " + bm:t")
}
form <- as.formula(modelbase)

# Fit with nlme::lme and AR(1) residual correlation
fit <- tryCatch(
  nlme::lme(form, random = ~1|ptID,
```

```
    correlation = nlme::corCAR1(form = ~t|ptID),
    data = datamerged,
    control = nlme::lmeControl(
      opt = "optim",
      maxIter = 200, msMaxIter = 200)),
  error = function(e) {
    # Fallback: no correlation structure
    tryCatch(
      nlme::lme(form, random = ~1|ptID,
        data = datamerged,
        control = nlme::lmeControl(opt = "optim")),
      error = function(e2) NULL
    )
  }
)

# Extract from nlme tTable
c <- summary(fit)$tTable
p <- c[target, "p-value"]
beta <- c[target, "Value"]
betaSE <- c[target, "Std.Error"]
```

6 Validation

The 200-replicate simulation confirms that the `corCAR1` model corrects the Type I error inflation:

Table 2: Type I error rates at $c_{bm} = 0$, no censoring, carryover = 0 (200 replicates). The `corCAR1` model restores rates to the nominal 5% range.

	`lmer` (before)		`lme + corCAR1` (after)	
Design	$N = 35$	$N = 70$	$N = 35$	$N = 70$
OL	0.13	0.09	**0.03**	**0.02**
OL+BDC	0.09	0.05	**0.03**	**0.04**
CO	0.17	0.13	**0.06**	**0.08**
N-of-1	0.06	0.08	**0.04**	**0.06**

All designs are now within the expected range for 200 replicates (a 95% binomial confidence interval for $p = 0.05$ at $n = 200$ is $[0.02, 0.09]$).

Power estimates with carryover also show the expected pattern:

The paper's central finding is confirmed with correct Type I error control: N-of-1 and OL+BDC designs are sensitive to carryover (power drops of 0.28–0.41 at $t_{1/2} = 1.0$ week), while the CO design is more robust (drops of 0.08–0.10). The power drops are moderate and clinically interpretable at realistic half-lives, unlike the artifact-driven crash ($0.95 \rightarrow 0.18$) in the original publication at $t_{1/2} = 0.1$ weeks.

Table 3: Power at $c_{bm} = 0.5$, no censoring, with `lme + corCAR1` analysis model (200 replicates). $\lambda_{cor} = \ln 2/t_{1/2}$ (auto).[1]

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$	Carryover effect
OL	35	0.08	0.06	0.09	None (always on drug)
OL	70	0.12	0.14	0.15	None
OL+BDC	35	0.34	0.24	0.18	Moderate drop
OL+BDC	70	0.68	0.55	0.27	**Large drop**
CO	35	0.27	0.25	0.19	Small drop
CO	70	0.43	0.44	0.34	Small drop
N-of-1	35	0.53	0.33	0.23	**Large drop**
N-of-1	70	0.83	0.74	0.55	**Large drop**

7 Summary of Changes Across the Analysis Model History

Table 4: Evolution of the analysis model

Version	Package	Correlation	Issue
Original (42ac030)	`lme4::lmer`	None (RI only)	Correct under CS
Ron Thomas (8609f12)	`lme4::lmer`	None (RI only)	Correct under CS
HEAD (pre-fix)	`lme4::lmer`	None (RI only)	Inflated Type I under AR(1)
HEAD (current)	`nlme::lme`	`corCAR1`	Correct under AR(1)

The transition from `lmer` to `lme` is driven entirely by the need to specify an explicit residual correlation structure, which `lme4` does not support. The fixed-effects formula is unchanged. The random effects structure (random intercept per participant) is unchanged. The only addition is the `corCAR1` term that accounts for the time-dependent residual autocorrelation introduced by the AR(1) DGP.

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